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Reshaping Corporate Boards Through Mandatory Gender Diversity Disclosures: Evidence from Canada

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Abstract. We study whether and how standardized gender diversity disclosure requirements affect women's representation on corporate boards. Our analysis takes advantage of Canada's 2014 regulation, which requires disclosures of consistent and comparable information on boards' gender diversity policies and guidelines. Using a difference-in-differences design, we find that the disclosure regulation increases women's representation on boards. The increase is more pronounced among firms with stronger shareholder oversight as proxied by ownership from environmental, social, and governance (ESG)–friendly investors, poor diversity performance preregulation, Toronto Stock Exchange index membership, and U.S. cross-listing. In addition, after the regulation, firms that make stronger diversity commitments (e.g., disclosing the adoption of written diversity policies and diversity targets) experience a higher increase in ownership by foreign ESG-friendly investors. Together, our findings suggest that standardized disclosure requirements enable shareholders to be better monitors, thereby improving board gender balance.

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Keywords: boards of directors • gender diversity • mandatory disclosure • shareholder oversight • corporate governance

1. Introduction

The growing global awareness of social equity has triggered an unprecedented wave of board gender diversity reforms worldwide (Deloitte 2017; Catalyst 2018a, b). An increasingly popular regulatory instrument to address gender imbalance on corporate boards is mandatory diversity disclosures, which aim to increase transparency to facilitate shareholder oversight. Critics argue that a lack of standardized disclosure requirements makes these mandates ineffective (Aguilar 2010, European Commission 2021). Yet we know little about whether and how standardized disclosure requirements improve women's representation on corporate boards. Our study addresses this question by taking advantage of Canada's 2014 regulation that requires companies to report standardized information on governance practices regarding gender diversity.

The 2014 regulation in Canada requires all domestic firms listed on the Toronto Stock Exchange (TSX) to disclose board and senior management gender-diversity policies and guidelines in their annual proxy circulars (Amendments to National Instrument 58-101, hereafter, the disclosure regulation). The disclosure regulation is "intended to provide investors and other stakeholders with information on the issuer's approach to advancing the representation of women on boards and in senior management, which in turn may impact investment and voting decisions" (Ontario Securities Commission 2014, p. 11). A unique feature of this regulation is its standardized disclosure requirements:[1] firms are required to disclose specific policies and guidelines (hereafter, diversity policies), including director term limits or other renewal mechanisms, written policies on the identification and nomination of women directors, consideration of women on the board, targets for women on the board, and the number and percentage of women on the board. The new information mandated by the regulation pertains to the board's diversity policies as information on the number of women directors was already available in the proxy filings prior to the regulation.

Canada's disclosure regulation provides a powerful setting to examine our research question. It does not require firms to adopt any diversity policy. Instead, it requires firms to explicitly disclose whether they have adopted specific policies and, if not, explain why not. In addition, the disclosure regulation is not bundled with governance code amendments or quota laws that recommend or require women's representation on the board. Furthermore, all TSX-listed issuers are subject to annual disclosure reviews by the securities regulators, which ensures the credibility of the disclosed information.

We posit that the disclosure regulation affects women's representation on boards by enabling shareholders to be better monitors. Shareholder engagement is costly and suffers from a free-rider problem (Grossman and Hart 1980). Before the disclosure regulation, it is difficult for investors to separate two types of firms: (1) firms with low diversity and no policies, referred to as "bad" firms, and (2) firms with low diversity but having a policy in place, referred to as "good" firms. Voluntary disclosure of diversity policies does not resolve the information problem because such disclosure often lacks credibility, verifiability, and standardization, especially when there is no well-accepted disclosure framework. Without consistent and comparable information, investors are unable to separate these two types of firms, which prevents them from targeting firms without policies and monitoring the progress at firms with policies.

The reporting obligation under the disclosure regulation forces firms to provide standardized information on whether they adopt any diversity policies, and if yes, what specific policies they adopt regardless of their performance and preferences. This information enables shareholders to identify and sort bad firms from good firms. Using the disclosed information, prosocial shareholders may cast unfavorable votes in director elections against bad firms, increasing ownership of good firms (Dimson et al. 2015, Dyck et al. 2019, Billings et al. 2022). Consistent with shareholder oversight as a catalyst for changes, a major investor put forward a shareholder proposal that asked Restaurants Brands International, which has an all-men board and owns Tim Horton, to adopt a written board-diversity policy following the disclosure regulation (Middlemiss 2016). To the extent that the anticipation and/or the actual impact of greater shareholder oversight increases the net benefits of improving board gender diversity, firms disclose the adoption of diversity policies to signal their good type. Because such public disclosures in a mandatory reporting regime bind firms to improving board diversity (diversity commitment), we hypothesize that firms increase women's representation on their boards following the disclosure regulation.

There are, however, arguments against finding an improvement in board gender diversity following the disclosure regulation. Shareholders may find the existing diversity information sufficient or perceive the disclosure of diversity policies as cheap talk. In addition, firms may not increase board diversity postregulation because the benefit from adopting a policy and improving diversity does not overweigh the additional cost of expanded recruitment effort and the reduced private benefits of incumbent directors even at the risk of being a target of activist shareholders. As men and women bring different skills to the board (Kim and Starks 2016), firms requiring more "male-oriented" skills may find it difficult to recruit women directors with these skills. Further, changing board composition is inherently a slow process because of the general lack of board turnover (Jiang 2023). Consequently, the disclosure regulation may not yield a discernable impact on board gender imbalance.

We test our hypothesis by examining changes in women's representation on the board in the four years before and after the regulation. We use a difference-in-differences (DiD) design that compares the postregulation change for Canadian TSX-listed firms (treatment firms) with the corresponding change for a benchmark sample of U.S. firms matched by an entropy balancing (EB) technique (EB U.S. firms). We find that the percentage of women directors increases by 53% after the disclosure regulation for the treatment firms relative to the benchmark firms, which indicates an increase from 0.869 to 1.33 in the average number of women directors. The magnitude of the increase is comparable to the 50% growth in women's representation on boards following the 2017 campaigns by the Big Three institutional investors (BlackRock, State Street, and Vanguard) in Gormley et al. (2023). Consistent with the parallel trends assumption, we find that, whereas the treatment and benchmark firms exhibit similar trends in the percentage of women directors preregulation, the treatment firms begin to increase women directors after the regulation relative to the benchmark firms. Our inference remains unchanged to excluding firms that voluntarily disclose their diversity practices before the regulation,[2] using alternative benchmarks (i.e., propensity score matched (PSM) U.S. firms, industry- and size-matched U.S. firms, non-U.S. firms, and Canadian venture firms) and using alternative dependent variables (e.g., the number of women directors and an indicator for the presence of a woman director).

Our cross-sectional analyses show that the increase in the percentage of women directors is larger when a firm faces stronger shareholder oversight. We capture shareholder oversight using several proxies measured in the year prior to the regulation: ownership from environmental, social, and governance (ESG)–friendly investors (i.e., signatories to the United Nations (UN)–supported Principles for Responsible Investment (PRI), pension funds and independent institutions, and the Big Three institutional investors), poor diversity performance, TSX index membership, and U.S. cross-listing. These results are consistent with the notion that firms anticipate enhanced oversight by prosocial investors and cater to them by adopting diversity policies and increasing diversity.

We also explore the influence of new women and men directors after the disclosure regulation. We find that the newly appointed women (men) directors are more (less) likely to chair major committees (i.e., governance, audit, compensation, and nominating committees). These results suggest that Canada's disclosure regulation improves women's influence on corporate boards.

Our next analysis assesses the link between diversity commitments and women's representation on boards. We measure diversity commitments using an index that sums up four indicators, which equal one (zero) if, in a given year after the regulation, a treatment firm discloses the adoption (nonadoption) of (1) director term limits and other renewal mechanisms, (2) a written policy regarding the representation of women on the board, (3) consideration of the representation of women in director recruitment, and (4) targets regarding the representation of women on the board. We find no difference in changes in women's representation on the board following the regulation between the EB U.S. Benchmark firms and treatment firm-years without disclosing any of the four diversity commitments. In addition, the post-regulation increase in women directors is larger for firm-years with stronger diversity commitments.

Our final set of analyses examines whether firms with diversity commitments gain more investment from ESG-friendly investors. We find that, after the regulation, firms with stronger diversity commitments experience a greater increase in the ownership by ESG-friendly investors that tend to respond more to disclosure regulations (i.e., UN PRI investors and foreign pension and independent institutions). These results suggest that the anticipated oversight facilitated by the disclosure regulation materializes and is commensurate with firms' diversity commitments. Lastly, we find no evidence supporting that government pressure (via government contracts) drives firms to increase board gender diversity.

Our study contributes to the literature in several ways. First, we extend the growing literature on how mandatory nonfinancial disclosure motivates firms to change their behavior (Christensen et al. 2017, Chen et al. 2018, Grewal et al. 2019, Downar et al. 2021). These disclosure regulations aim to alter firm behavior by affecting the cost–benefit trade-off through improved transparency rather than by directly stipulating certain firm behavior. In the classic disclosure–real effect setting, mandatory reporting of a firm's performance brings public scrutiny and incentivizes firms to change their behavior. For example, Christensen et al. (2017) find that mine safety improved following the mandatory disclosure of mine-safety records in the financial reports of U.S. Securities and Exchange Commission (SEC)–registered mine owners. Downar et al. (2021) document that UK firms decreased direct carbon emissions following the mandatory disclosure of carbon emissions in their annual reports. In contrast, the performance measure (i.e., board gender composition) in our setting was already known prior to the regulation. As such, the mechanism we document differs; the required disclosure of a firm's diversity policies enables shareholders to better monitor boards' diversity policies and progress, thereby changing firms' behavior in appointing women directors.

Second, we contribute to the literature on the effect of corporate gender diversity reforms and board structures.[3] Most of this research focuses on gender quota laws and comply-or-explain governance code amendments. Using the 2003 gender quota reform in Norway, Ahern and Dittmar (2012) and Matsa and Miller (2013) find that firm value and financial performance decrease following the reforms, whereas Eckbo et al. (2022) find no effect. Using governance code reforms that aim to improve gender diversity on boards from 25 countries, Fauver et al. (2024) suggest that these reforms empower socially conscious foreign institutional investors to drive value-enhancing governance changes. Whereas mandatory diversity disclosure has gained popularity, evidence on its effectiveness is scant.[4] Our evidence is consistent with the Canadian disclosure requirement, in fact, moving the needle dramatically on women's representation on boards.

Third, our findings speak to the implementation and design of nonfinancial disclosure regulations. Unlike disclosure regulations on financial reporting and internal control systems (Doyle et al. 2007, DeFond et al. 2011), nonfinancial disclosure regulations typically allow multiple reporting frameworks and lack consistent requirements (European Commission 2021). In contrast to the 2009 U.S. diversity disclosure rule that is generally perceived as ineffective, Canada's 2014 regulation prescribes standardized reporting requirements on gender-diversity policies. Our findings not only help assess whether Canada's regulation achieves its objective "to assist investors when making investment and voting decisions" (Ontario Securities Commission 2014, p. 4) but also provide implications for the implementation of global nonfinancial disclosure regulations (European Commission 2021, U.S. Securities and Exchange Commission 2024).

Finally, our study extends the extensive work on shareholder activism (see Denes et al. 2017 for a review). A growing stream of literature investigates investor engagement on ESG issues (Dimson et al. 2015, Dyck et al. 2019, Azar et al. 2021, Gormley et al. 2023). We add to this research by providing evidence suggesting that standardized diversity information makes shareholders a better monitor and that shareholders value the disclosed commitment in their investment decisions.

2. Institutional Background and Hypothesis Development

2.1. Institutional Background

Over the past few decades, women's participation in the labor market has grown substantially in Canada. According to Statistics Canada (2011), women make up 47% of the Canadian workforce. As of 2011, however, women directors at public companies accounted for only 10%, and about 40% of companies had no woman on boards (Catalyst 2012). In June 2013, the Ontario Minister of Finance requested that the Ontario Securities Commission (OSC) undertake a public consultation regarding disclosure requirements for women on boards and in senior management. The OSC published a consultation paper in July 2013 and convened a public roundtable to discuss the proposed requirements in October 2013. In January 2014, after receiving more than 90 written submissions to the consultation paper and more than 440 responses to a subsequent survey, the OSC issued for public comment proposed Amendments to National Instrument 58-101 (Disclosure of Corporate Governance Practices).[5] On October 15, 2014, the OSC and seven other securities regulators issued the Multilateral CSA Notice of Amendments to National Instrument 58-101. Whereas the disclosure regulation is not adopted by all Canadian Securities Administrators (CSA) members, TSX-listed companies are subject to the disclosure requirements as they are reporting issuers in Ontario. The new requirements apply to domestic issuers listed on the TSX following the fiscal year ending on or after December 31, 2014, but not to those listed on the TSX Venture Exchange.

The objectives of the regulation are "to encourage more effective boards and better corporate decision making by requiring greater transparency for investors and other stakeholders … This transparency is intended to assist investors when making investment and voting decisions" (Ontario Securities Commission 2014, p. 6). The regulation adds the following items in Form 58-101F1 Corporate Governance Disclosure: (1) item 10, director term limits and other mechanisms of board renewal; (2) item 11, policies regarding the representation of women on the board; (3) item 12, consideration of the representation of women in the director identification and selection process; (4) item 13, consideration given to the representation of women in executive officer appointments; (5) item 14, the issuer's targets regarding the representation of women on the board and in executive officer positions; and (6) item 15, number of women on the board and in executive officer positions.[6] For example, item 11 requires firms to "disclose whether the issuer has adopted a written policy relating to the identification and nomination of women directors. If the issuer has not adopted such a policy, disclose why it has not done so." We focus on the disclosure requirements regarding women on boards because the requirements regarding women in executive positions are relatively limited and unspecific (e.g., no disclosure requirement of policies regarding women executives).

The 2014 Canadian regulation offers several advantages for studying our research question. First, there is no requirement for firms to adopt any specific diversity policy. The likelihood of quota-based regulation is small because the Canadian system of corporate governance regulation features mandatory disclosure of either the adoption of certain governance practices or an explanation of alternative practices. The OSC consultation paper notes that "board diversity is not about quotas or tokenism" (Ontario Securities Commission 2013, p. 6). Second, Canada's disclosure regulation does not accompany governance code amendments or quota legislation, thereby allowing us to isolate the effect of disclosure from other concurrent regulatory reforms. Third, the regulators conduct annual disclosure reviews for all TSX-listed issuers. The review reports and associated data are publicly available on the OSC's website, thereby ensuring information transparency and credibility.[7]

Appendix A provides sample disclosures from Air Canada and Microbix Biosystems, which have a diversity commitment index of four and zero, respectively, in the first year after the disclosure regulation. Neither company provides any discussion related to diversity in their proxy circulars of the year prior to the regulation (year -1). However, Air Canada increases the number of women directors from one (in year 0) to three (in year 2), whereas Microbix Biosystems remains with an all-men board.

Because Canada's regulation aims to address the weaknesses of diversity disclosure rules in other countries, it is worth comparing this regulation with the U.S. rule. In 2009, the SEC amended Regulation S-K to require firms to provide information on how diversity in director nominations is considered and implemented. Because the U.S. rule does not define diversity, nor does it require a firm to disclose the nonadoption of a written diversity policy, companies "often times [limit] their disclosure to a brief statement indicating [that] diversity was something considered as part of an informal policy. Many companies did not include any discussion of any concrete steps taken to give real meaning to its efforts to create a diverse board" (Aguilar 2010).[8]

2.2. Hypothesis Development

We posit that Canada's 2014 disclosure regulation provides information that enables shareholders to be better monitors, which leads to improved board gender diversity. Prior to the regulation, there exist significant information frictions that prevent shareholders from effectively separating bad firms (e.g., firms with low diversity and no diversity policies) from good firms (e.g., firms with low diversity but having a policy in place). Prior studies suggest that full disclosure is rare

(Verrecchia 1983) in a voluntary disclosure regime, and when voluntary disclosures occur, they often lack credibility, verification, and standardization (She 2022). Firms may simply say that they have an informal policy that considers diversity. Because of the lack of consistent and comparable information of firms' diversity policies to differentiate between good firms and bad firms, investors are unable to identify the bad firms to exert pressure or to monitor the diversity progress of the good firms.

The regulation requires disclosure of standardized diversity information, which allows investors to easily separate bad firms from good firms and, hence, facilitates their benchmarking and monitoring. Based on the disclosed information, prosocial shareholders may voice their disapproval by casting unfavorable votes in director elections against bad firms or show their support by increasing ownership of good firms. In anticipation of the improved oversight, firms adopt diversity policies to signal their good type to prosocial investors. These disclosures serve as diversity commitments that bind firms to improving board diversity because diversity policies often involve establishing responsible parties, devising means, and evaluating progress (Sull 2003). We, therefore, expect more firms to adopt diversity policies and improve board gender diversity following the disclosure regulation.

Consistent with the notion that the disclosure regulation facilitates shareholder engagement on diversity practices, anecdotal evidence suggests that shareholders regard the regulation as a catalyst for change. For example, in response to the all-men board of Tim Horton's parent company (Restaurants Brands International) in 2016, a key investor presented a shareholder proposal at the annual meeting that asks the company to adopt a written board-diversity policy.[9] Appendix B provides an excerpt of the Canada Proxy Voting Policy Update and Guidelines by the Institutional Shareholder Services (ISS). In its 2017 policy update, the ISS states, "Gender diversity has become a high-profile corporate governance concern in the Canadian market since updated disclosure requirements came into effect for the 2015 proxy season." In its 2018 Proxy Voting Guidelines, the ISS includes commitments to board gender diversity (based on the National Instrument 58-101 disclosures) as part of the recommended voting guideline on director nominees for all companies listed on the TSX. For companies in the S&P/TSX Composite Index, it further recommends withholding votes for the chair of the nominating committee when a company has not disclosed a formal written gender-diversity policy and has zero women directors.

In sum, we posit that Canada's disclosure regulation provides useful information that facilitates shareholder monitoring. The prospect of increased shareholder monitoring incentivizes firms to adopt a gender-diversity policy and improve board gender diversity. Our hypothesis follows.

Hypothesis 1. *Firms increase women's representation on their boards following Canada's 2014 gender diversity disclosure regulation.*

There are several arguments for why the disclosure regulation may yield little impact. First, disclosure regulations may lack teeth in promoting corporate gender diversity. It is also unclear whether standardized disclosure requirements on diversity practices have incremental value because information on the number of women directors was already available in the proxy filings before the regulation. Second, appointing women directors may involve significant search costs. To the extent that men and women bring different skills to the board (Kim and Starks 2016), firms requiring more male-oriented skills may find it challenging to find women directors with these skills. Adding women directors can also face resistance from incumbents who prefer to appoint directors from the existing executive networks, particularly when firms have low visibility or face little public scrutiny. Firms, in anticipation of shareholder monitoring of diversity policies following the disclosure regulation, trade off the benefit from shareholder support and potential improvements in board effectiveness vis-à-vis the loss of their private benefits and the expanded recruitment efforts. To the extent that the cost outweighs the benefit, firms may remain a nonadopter of diversity policies even at the risk of being a target of shareholder campaigns after the regulation. Lastly, change in board composition is inherently a slow process because incumbents generally enjoy long tenure.

3. Data and Research Design

3.1. Sample

Our treatment firms consist of Canadian domestic firms listed on the TSX that are subject to the disclosure regulation. We restrict the sample period to the four years before Canada enacted the regulation in December 2014 and the four years after. The preregulation period consists of the fiscal years ending in December 2010 through November 2014, and the postregulation period consists of the fiscal years ending in December 2015 through November 2019. We exclude the effective year of the regulation (year 0), that is, the year ending in December 2014 through November 2015, because we cannot assess the relation between women directorships in year 0 and the lagged diversity commitment (unavailable in year -1).

We obtain financial data from S&P Compustat, director profiles from BoardEx, and institutional holding data from the Thomson Reuters Global Ownership database. For our primary analyses, we require sample firms to have the necessary financial and director data and to have at least one year in both preregulation and postregulation periods. These criteria yield a treatment sample of 2,339 firm-year observations for 381 unique firms.[10] Table 1 presents the sample selection procedure (panel A) and distribution by year (panel B) and industry (panel C). Forty-seven and nine tenths percent (381/796) of the TSX-listed firms are included in our final sample. As shown in Online Table A1, the industry composition of our final sample is comparable to that of the initial sample.

To perform DiD analyses, we use an entropy balancing technique (Hainmueller and Xu 2013, McMullin and Schonberger 2020) to construct a benchmark sample of matched U.S. firms (EB U.S. sample). Relative to other matching approaches, an advantage of entropy balancing is that it eliminates differences in observable covariates across treatment and benchmark samples without reducing the sample size. By matching on institutional ownership and governance characteristics, this approach helps ensure that our treatment and benchmark firms have similar incentives to improve governance. We perform entropy balancing on the first and second moments of covariates and set the tolerance level at 0.015. The EB U.S. sample consists of 19,559 firm-years for 3,026 unique U.S. firms. Table 1, panels B and C, presents the distribution of the EB U.S. sample by year and industry. All but two industries in the treatment sample (i.e., nonmetallic minerals except fuels and radio and TV broadcasting) experience an increase in women's representation on boards after the regulation. Compared with the EB U.S. firms, all industries in the treatment sample experience a relative increase except for five industries.

3.2. Research Design

We test our hypothesis using a DiD design that compares women's representation on corporate boards before and after the regulation for treatment versus benchmark firms. Specifically, we regress the percentage of women directors on an interaction term between a variable indicating the postregulation period (*Post*) and a variable indicating the treatment firms (*Treat*), an array of control variables, and firm and year fixed effects. To ease comparison with prior research (Gormley et al. 2023, Fauver et al. 2024), we use the percentage of women directors as our primary measure. We omit the indicator variables, *Post* and *Treat*, because they are absorbed by the year and firm fixed effects. Our regression model follows:

$$\begin{aligned}\%\ \mathit{women\ directors}_{t+1} = \beta_0 + \beta_1 \mathit{Post} \times \mathit{Treat} + \mathit{Controls}_t \\ + \mathit{Firm\ FEs} + \mathit{Year\ FEs} + \mu.\end{aligned} \quad (1)$$

The coefficient on *Post* × *Treat* captures the postregulation change in the percentage of women directors for the treatment firms relative to that for the benchmark firms. We control for the following variables that likely explain firms' gender diversity performance (Gul et al. 2011, Knippen et al. 2019): (1) *Inst. ownership*, measured as total number of shares owned by institutional investors divided by shares outstanding; (2) *Woman CEO*, a variable indicating a woman CEO; (3) *CEO-chair*, a variable indicating whether the firm's CEO is the board chairperson; (4) *Indp. directors*, the proportion of independent directors; (5) *Avg. dir. tenure*, the natural logarithm of one plus the average tenure of a firm's directors; (6) *Board size*, the natural logarithm of the number of directors; (7) *Market-book*, the ratio of the market value of equity to the book value of equity; (8) *Leverage*, short-term debt plus long-term debt divided by total assets; (9) *Firm size*, the natural logarithm of total assets in millions of U.S. dollars; (10) *ROA*, pretax income divided by total assets; (11) *Ret.*, annual stock returns adjusted by the industry return; and (12) *Firm age*, the natural logarithm of firm age. All control variables are lagged by one year. We use robust standard errors clustered by firm to evaluate the significance of the regression coefficients in all analyses.[11]

3.3. Descriptive Statistics

Table 2, panel A, reports summary statistics for the treatment and EB U.S. Benchmark samples. The average percentage of women directors (*% women directors*), number of women directors (*N. women directors*), and percentage of firms having women directors (*Having women directors*) over the sample period are 11.62%, 1.199, and 61.8% for the treatment firms and are 10.41%, 1.024, and 61.4% for the EB U.S. firms, respectively. For the control variables, the treatment and EB U.S. samples exhibit similar means and standard deviations as we perform the entropy balancing on the first and second moments of covariates. Panel B reports results of the univariate analysis. Comparing changes in women's representation on the board from the preregulation to postregulation period, the increase in the percentage of women directors for the treatment firms is significantly greater than that for the EB U.S. firms. The average percentage of women directors for the treatment firms increases from 7.88% to 14.93%, whereas the corresponding increase is from 8.88% to 12.25% for the EB U.S. firms. The DiD increase is 3.68%, significant at the 1% level.

4. Empirical Analysis

4.1. Gender Diversity Disclosure Regulation and Women's Representation on Boards

Table 3 presents the results of testing our hypothesis. Columns (1) and (2) report the results using the treatment firms only with and without including the control variables, respectively. The coefficient on *Post* is positive and significant in both columns, suggesting that the

Table 1. Sample Selection and Distribution

Panel A: Sample selection procedure for treatment firms

Selection criteria for treatment firms	Number of Firm-years	Number of Firms
TSX-listed Canadian domestic firms with nonmissing financial and stock price data from Compustat North America, fiscal year ending in December 2009 to November 2018 (initial sample)	5,166	796
Appear at least one year in both preperiod and postperiod in Compustat North America	4,648	607
Covered by BoardEx	3,358	600
Appear at least one year in both preperiod and postperiod in BoardEx	2,677	381
Excluding year 0 (final sample)	2,339	381

Panel B: Sample distribution by event year

Year	Treatment sample	EB U.S. sample
Preregulation		
Year −4 (Dec. 2010–Nov. 2011)	213	2,258
Year −3 (Dec. 2011–Nov. 2012)	240	2,419
Year −2 (Dec. 2012–Nov. 2013)	301	2,608
Year −1 (Dec. 2013–Nov. 2014)	346	2,687
Postregulation		
Year +1 (Dec. 2015–Nov. 2016)	354	2,757
Year +2 (Dec. 2016–Nov. 2017)	326	2,439
Year +3 (Dec. 2017–Nov. 2018)	291	2,293
Year +4 (Dec. 2018–Nov. 2019)	268	2,098
Total	2,339	19,559

Panel C: Percentage of women directors by industry

	Treatment firms					EB U.S. firms					
	N	Year −1 (a)	Year +1 (b)	Years 1–4, mean (c)	Years 1–4 versus −1 (c–a)	N	Year −1 (d)	Year +1 (e)	Years 1–4, mean (f)	Years 1–4 versus −1 (f–d)	DiD, years 1–4 versus −1 ((c–a)–(f–d))
Mining	616	5.4	9.4	11.3	5.9	193	2.7	2.2	2.4	−0.4	6.2
Oil & gas extraction	377	3.3	5.8	8.8	5.5	795	4.6	6.4	7.6	3.0	2.5
Nonmetalic minerals except fuels	22	7.5	2.3	7.2	−0.4	39	0.8	3.3	1.3	0.6	−1.0
Construction	21	10.3	11.1	17.0	6.8	250	1.5	5.1	7.8	6.4	0.4
Food & kindred products	43	10.3	17.4	20.0	9.7	406	14.8	15.0	14.7	−0.1	9.8
Lumber & wood products	52	6.2	8.9	12.1	5.9	96	4.4	7.2	11.4	7.0	−1.2
Printing & publishing	23	19.1	25.8	23.3	4.2	109	11.1	12.3	21.0	9.9	−5.7
Chemicals & allied products	146	6.4	10.4	12.7	6.3	1,821	8.0	11.8	12.5	4.5	1.8
Petroleum & coal products	31	15.8	18.3	21.6	5.8	106	5.0	7.3	15.3	10.3	−4.5
Machinery, except electrical	29	6.6	10.4	15.0	8.4	947	7.2	7.2	13.1	5.9	2.5
Electrical & electronic equipment	70	8.1	13.5	15.3	7.2	1,332	8.9	10.7	11.2	2.3	4.9
Transportation equipment	39	9.5	18.6	19.4	9.9	444	8.5	8.4	13.6	5.1	4.9
Instruments & related products	23	2.8	5.6	9.1	6.3	1,105	9.8	8.9	11.0	1.2	5.1
Transportation	68	11.5	16.9	22.6	11.1	507	6.1	5.2	7.0	0.8	10.3
Telephone & telegraph comm.	24	14.2	24.2	26.8	12.5	140	20.6	16.1	15.3	−5.3	17.8
Radio & television broadcasting	21	30.1	22.7	26.4	−3.8	303	9.0	9.3	8.9	−0.1	−3.7
Electric, gas, & water supply	72	19.1	25.7	30.1	11.0	664	16.3	18.5	19.4	3.1	7.9
Wholesale	64	8.8	13.7	18.4	9.6	607	8.8	8.9	14.3	5.4	4.1
Retail stores	79	13.4	20.7	20.1	6.7	896	14.1	17.0	17.0	2.9	3.8
Finance, insurance, & real estates	248	10.4	15.7	18.2	7.8	4,355	9.8	11.2	12.3	2.5	5.3
Services	164	10.2	17.5	20.7	10.5	2,951	9.4	11.1	12.2	2.8	7.8
Others	107	7.4	12.5	17.5	10.1	1,493	7.5	9.3	12.0	4.6	5.5
Total	2,339	7.8	12.0	14.9	7.1	19,559	9.5	10.9	12.3	2.8	4.3

Notes. This table presents sample selection and distribution. Panel A reports the sample selection procedure for treatment firms. Panel B reports the sample distribution by year. Panel C reports the percentage of women directors for the treatment sample and the EB U.S. sample by Fama–French 38 industry classification, in which industries with less than 10 treatment firm-year observations are combined into "Others." See Appendix C for variable definitions.

percentage of women directors increases for the treatment firms after the disclosure regulation. In columns (3) and (4), we estimate Equation (1) after including the EB U.S. Benchmark sample. The coefficient on *Post* × *Treat* is positive and significant in both columns, suggesting a greater increase in women's representation among the treatment firms relative to the benchmark firms. The increase in women directors is economically meaningful. Based on column (4), the average percentage of women directors increases by 53% relative to the EB U.S. firms following the disclosure regulation, which translates to an increase from 0.869 to 1.33 in the average number of women directors.[12] This increase is comparable to the 50% increase in women representation on the board following the 2017 Big Three campaign documented in Gormley et al. (2023).

In column (5) of Table 3, we perform a dynamic analysis to test the parallel trends assumption underlying our DiD design. We reestimate Equation (1) by replacing *Post* with seven year indicators (i.e., *Year −4*, *Year −3*, *Year −2*, *Year 1*, *Year 2*, *Year 3*, and *Year 4*), and set *Year −1* as the base year. We find that, whereas the coefficients on *Treat* × *Year −4, Treat* × *Year −3*, and *Treat* × *Year −2* are insignificant at conventional levels, those on *Treat* × *Year 1*, *Treat* × *Year 2*, *Treat* × *Year 3*, and *Treat* × *Year 4* are positive and significant. Figure 1 plots the coefficients to illustrate the parallel trends and shows that the relative increase in the percentage of women directors for the treatment firms occurs only after the regulation.

Regarding the control variables, firms with a woman CEO (*Woman CEO*), a larger board size (*Board size*), and higher growth (*Market-book*) have a greater percentage of women directors. Despite differences in the sample and period, these findings are generally consistent with those in prior studies (e.g., Gul et al. 2011, Knippen et al. 2019). Collectively, the results in Table 3 are consistent with our hypothesis and suggest that the disclosure regulation in Canada leads to an increase in women's representation on corporate boards.

4.2. Robustness Checks

We conduct a series of tests to check the robustness of our main results in columns (2) and (4) of Table 3. Table 4 reports the results.

4.2.1. Alternative Treatment Samples and Event Windows. We note that our story only requires that shareholders obtain credible disclosures postregulation regarding a firm's diversity policies for benchmarking and monitoring. It does not matter whether a firm had no policy or already had a policy internally but did not disclose it prior to the regulation.[13] Nevertheless, a potential concern for our inference is that firms may already disclose their diversity policies under the voluntary disclosure regime. To shed light on this issue, we hand-collect information from proxy circulars in the Canadian System for Electronic Document Analysis and Retrieval database on voluntary gender diversity disclosures in year −1. Whereas the proposed disclosure requirements were already released in July 2013, we find that only 72 (21.6%) out of the 334 treatment firms with available proxy circulars in year −1 provide disclosures related to gender diversity in their proxy circulars. The disclosures of these firms tend to be vague and vary in scope. Only 27 (8.1%) disclose the adoption of director term limits, 16 (4.8%) disclose the adoption of a board-diversity policy, 64 (19.2%) disclose considerations of women's representation in director recruitment, and 8 (2.4%) disclose the adoption of a gender-diversity target.

We include firms that voluntarily disclose their policies throughout our analysis because these firms can still change their disclosures and strengthen their diversity commitments. A firm, for example, may expand its disclosure from reporting only the consideration of gender diversity in director recruitment preregulation to further including the adoption of a written diversity policy postregulation. To rule out the potential confounding effect of a firm's voluntary diversity disclosures, we rerun our analyses after excluding the firms with voluntary disclosures of diversity policies in year −1 in columns (1) and (2) of Table 4, panel A. In columns (3) and (4), we rerun our analyses after restricting the sample to an alternative event window of three years before and after the disclosure regulation. Our results remain robust.[14]

4.2.2. Alternative Benchmark Samples. Our primary analyses use the EB U.S. firms as the benchmark sample. In this section, we construct four alternative benchmark samples for robustness checks. Online Table A2 reports the sample selection for the alternative benchmark samples. The first is propensity score matched U.S. firms (PSM U.S. sample). As described in detail in Online Table A3, the PSM procedure generates 297 matched pairs with 1,964 U.S. firm-years. To mitigate the concern of our results being driven by industry differences between the treatment and benchmark samples, we construct the second alternative benchmark sample by matching a treatment firm with a U.S. firm in the same industry with the closest market capitalization in year −1 (Industry- & size-matched U.S. sample of 2,485 firm-years for 381 U.S. firms). The third alternative benchmark sample is non-U.S. firms from economies without legislation-based or governance-code board gender-diversity reforms (Catalyst 2018a, b; Fauver et al. 2024) (non-U.S. sample). After excluding economies with fewer than 50 firm-year observations, the non-U.S. sample consists of 1,553 observations for 286 firms from eight economies: Argentina, Brazil, China, Indonesia, South Korea, Mexico, New Zealand, and Philippines.

Table 2. Descriptive Statistics

Panel A: Summary statistics

Variable	Treatment sample (observations = 2,339 firm-years)			EB U.S. sample (observations = 19,559 firm-years)		
	Mean	Median	Standard deviation	Mean	Median	Standard deviation
% women directors	11.62	11.11	11.36	10.41	10.00	10.38
N. women directors	1.199	1.000	1.340	1.024	1.000	1.076
Having women directors	0.618	1.000	0.486	0.614	1.000	0.487
Post	0.530	1.000	0.499	0.454	0.000	0.498
Inst. ownership	0.295	0.257	0.236	0.295	0.280	0.236
Woman CEO	0.033	0.000	0.178	0.033	0.000	0.178
CEO-chair	0.178	0.000	0.383	0.178	0.000	0.383
Indp. directors	0.734	0.750	0.135	0.734	0.750	0.135
Avg. dir. tenure	2.030	2.071	0.453	2.030	2.075	0.453
Board size	2.165	2.197	0.303	2.165	2.197	0.303
Market-book	2.383	1.653	2.949	2.383	1.473	2.949
Leverage	0.195	0.175	0.173	0.195	0.145	0.173
Firm size	6.837	6.783	2.139	6.837	6.871	2.139
ROA	−0.026	0.023	0.245	−0.026	0.013	0.245
Ret	0.144	0.040	0.599	0.144	0.031	0.599
Firm age	3.023	2.944	0.769	3.023	2.996	0.769

Panel B: Univariate analysis

Variable = % *women directors*	Treatment firms	EB U.S. firms	Difference (*t*-statistics)
Preregulation period	7.88	8.88	−1.00*
			(−1.64)
Postregulation period	14.93	12.25	2.68***
			(3.94)
Change (post versus pre)	7.05***	3.37***	3.68***
(*t*-statistics)	(14.75)	(9.10)	(6.11)

Notes. This table presents summary statistics. Panel A presents descriptive statistics of variables for the treatment and EB U.S. samples. Panel B presents the univariate results comparing the percentage of women directors in the preregulation and postregulation periods across the treatment sample and the EB U.S. sample. See Appendix C for variable definitions.
***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Online Table A4 lists corporate governance requirements regarding board independence and audit committees for these eight countries and the United States. All of these non-U.S. countries, except for Argentina and Brazil, require board independence and an audit committee. The fourth alternative benchmark sample consists of Canadian firms listed on the TSX Venture Exchange (venture firms), which are exempt from the disclosure regulation. The sample of venture firms consists of 229 firm-year observations for 41 firms. Online Table A5 provides summary statistics for these alternative benchmark samples.

An advantage of using the PSM U.S. and industry- & size-matched U.S. samples as the benchmark is that these firms are more comparable to our treatment firms than the non-U.S. firms. A limitation, which also applies to the EB U.S. sample, is that the U.S. Council of Institutional Investors amended the Corporate Governance Policies to recommend consideration of gender in 2013, a potential confounding event during our sample period. The sample sizes of these alternative U.S. benchmarks are also smaller than our primary EB U.S. Benchmark. Canadian venture firms serve as a useful benchmark because they are subject to the same national institutions and concurrent political changes (e.g., the election of Justin Trudeau with a big win of the Liberal and Green Parties in 2015) that coincided with the increased stakeholder support of the gender diversity issues but are exempt from the disclosure regulation. However, the venture firms (almost equivalent of U.S. over-the-counter firms) differ substantially from treatment firms with a very high percentage delisting or failing. In panel B of Table 4, we rerun our analyses using the four alternative benchmark samples. Our results remain robust.

4.2.3. Alternative Dependent Variables. We rerun our analysis after using the number of women directors (*N. women directors*) and a variable indicating whether a firm has at least one woman director (*Having women directors*) as alternative dependent variables. As *N. women directors* is a count variable with many zero-value observations, we employ a zero-inflated Poisson model (Cohn et al. 2022). When the dependent variable is the indicator for women directors, we use a probit model.

Table 3. Gender Diversity Disclosure Regulation and Women's Representation on Boards

	Dependent variable = % *women directors*$_{t+1}$				
Sample =	Treatment firms only		Treatment & EB U.S. firms		
	(1)	(2)	(3)	(4)	(5)
Post	**7.430*****	**5.386*****			
	(15.49)	**(9.03)**			
Treat × Post			**4.151*****	**4.200*****	
			(7.38)	**(7.44)**	
Treat × Year −4					−0.202
					(−0.38)
Treat × Year −3					−0.127
					(−0.26)
Treat × Year −2					−0.224
					(−0.61)
Treat × Year 1					**2.422*****
					(4.86)
Treat × Year 2					**3.714*****
					(6.15)
Treat × Year 3					**4.909*****
					(7.19)
Treat × Year 4					**6.373*****
					(7.95)
Inst. ownership		3.078		1.950	1.700
		(1.49)		(1.61)	(1.38)
Woman CEO		3.898		6.448***	6.469***
		(1.64)		(4.73)	(4.74)
CEO-chair		0.044		0.442	0.384
		(0.04)		(0.76)	(0.66)
Indp. director		2.386		1.330	1.238
		(0.85)		(0.82)	(0.76)
Avg. dir. tenure		−1.763		−0.428	−0.335
		(−1.48)		(−0.63)	(−0.50)
Board size		2.805*		2.938***	2.997***
		(1.73)		(2.93)	(3.01)
Market-book		0.134		0.101*	0.111**
		(1.30)		(1.87)	(2.00)
Leverage		−3.294		−1.922	−2.100
		(−1.46)		(−1.35)	(−1.46)
Firm size		0.732		0.383	0.454
		(1.30)		(0.93)	(1.09)
ROA		0.701		0.292	0.177
		(0.66)		(0.40)	(0.24)
Ret		0.321		−0.188	−0.204
		(1.44)		(−1.23)	(−1.32)
Firm age		7.865***		0.861	0.711
		(3.72)		(0.62)	(0.51)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	No	No	Yes	Yes	Yes
Observations (firm-years)	2,339	2,339	21,898	21,898	21,898
Adjusted R^2	0.738	0.748	0.775	0.780	0.783

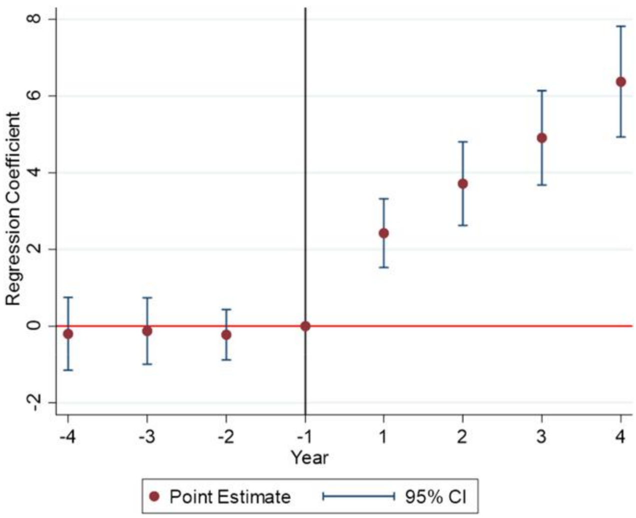
Notes. This table presents the results of the effects of Canada's gender diversity disclosure regulation on women's representation on boards. All firm-year independent variables are lagged by one year. See Appendix C for variable definitions. The *t*-statistics reported in parentheses are calculated based on robust standard errors clustered by firm.

***, **, and * indicate significance at the 1%, 5%, and 10% two-tailed levels, respectively.

We include *Treat* in the models after replacing firm fixed effects with industry fixed effects because nonlinear models with many fixed effects likely suffer from incidental parameter problems (Wooldridge 2010). As shown in columns (1) and (2) of Table 4, panel C, our results remain robust.

To gauge whether the increase in women directors is merely through adding more board seats, we examine the change in men directors and in board size after the disclosure regulation. In columns (3) and (4) of Table 4, panel C, we rerun our analysis using the number of men directors (*N. men directors*) and the number of directors

Figure 1. (Color online) Diagnostic Test of the Parallel Trends Assumption



Notes. This figure plots the coefficients on the interactions between *Treat* and the year indicators in the regressions reported in Table 3. The solid dots represent the point estimates, and the lines represent the 95% confidence intervals. Year −1 is set as the benchmark year.

(*N. directors*) as the dependent variable, respectively. Using a Poisson model, we find that the coefficient on *Post* × *Treat* is significantly negative, suggesting a reduction in the number of men directors following the disclosure regulation among the treatment firms relative to the benchmark firms. The Poisson model in column (4) further shows no significant change in board size after the regulation among the treatment firms relative to the benchmark firms, suggesting that firms add new women directors through the substitution of incumbent men directors.

5. Additional Analyses

5.1. Analyses Conditional on Shareholder Oversight

We argue that standardized disclosure requirements provide informational resources that facilitate shareholder oversight in promoting board gender diversity. In anticipation of the enhanced oversight, firms adopt diversity policies to cater to prosocial investors. To corroborate our arguments, we test whether firms exhibit a greater increase in women's representation on the board when they face stronger shareholder oversight. We use two sets of proxies to capture the extent of shareholder oversight, all measured in year −1: (1) proxies related to investor type, measured as ownership held by ESG-friendly institutional investors, including signatories to the UN PRI, pension funds and independent institutional investors, and the Big Three investors; and (2) other proxies, including a variable indicating whether the percentage of women directors is less than the industry median (*Underrep. of women dir.*), a variable indicating whether a firm is included in the TSX composite index (*TSX index*), and a variable indicating whether a firm is cross-listed on U.S. stock exchanges (*U.S. cross-listing*). The UN PRI is an organization whose goal is to advance the integration of ESG into investment analysis and decision making. Compared with hedge funds that have a relatively short investment horizon and primarily focus on financial returns, pension funds and independent institutions (including mutual funds and investment advisers) have a greater need to cater to their clients on social issues. Prior studies find that these investors, as well as the Big Three, tend to engage in ESG issues and possess greater influence on firms when they hold a larger position (Dyck et al. 2019, Gormley et al. 2023). We also expect that firms with poor gender diversity prior to the regulation are likely under greater scrutiny, and firms with TSX index membership and those with shares cross-listed in the U.S. are subject to more intensive shareholder involvement because of their visibility, larger shareholder base, and/or a higher burden for reputational bonding (Dimson et al. 2015).

Table 5, panel A, reports summary statistics of the shareholder oversight proxies. Panels B and C report the regression results using proxies related to investor type and other proxies. We perform the analyses only among the treatment firms when the proxies (i.e., TSX index and U.S. cross-listing) do not apply to the benchmark firms. We find that the coefficients on our variables of interest (*Post* × *Treat* × *UN PRI IO*, *Post* × *Treat* × *Pension & indp. IO*, *Post* × *Treat* × *Big 3 IO*, *Post* × *Treat* × *Underrep. of women dir.*, *Post* × *TSX index*, and *Post* × *U.S. cross-listing*) are all significantly positive. These results suggest that firms under greater ex ante shareholder oversight have a greater increase in women's representation on boards following the disclosure regulation.

5.2. Analysis of Committee Chairs and Board Chairs/Lead Directors

It is possible that the increase in women's representation on the board may simply reflect tokenism or window dressing rather than meaningful governance changes. To assess whether this increase is merely a symbolic way to maintain institutional legitimacy or manage impression (DiMaggio and Powell 1983, Knippen et al. 2019), we focus on the subsample of firm-years with a new director appointed and examine whether the newly appointed women directors are more likely to chair major board committees (*Woman committee chair, new directors*) and become the board chair or lead director (*Woman board chair/lead director, new directors*) after the regulation. In comparison, we also examine whether the newly appointed men directors are less likely to chair major board committees (*Man committee chair, new directors*) and become the board chair or lead director (*Man board chair/lead director, new directors*).

Table 4. Robustness Checks

Panel A: Alternative treatment samples and event windows				
Dependent variable =	*% women directors$_{t+1}$*			
	Excl. voluntary reporters		[−3, +3] window	
Sample =	Treatment firms only (1)	Treatment & EB U.S. firms (2)	Treatment firms only (3)	Treatment & EB U.S. firms (4)
Post	**4.750*****		**5.179*****	
	(6.76)		**(8.96)**	
Post × Treat		**3.912*****		**3.666*****
		(6.41)		**(6.62)**
Controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	No	Yes	No	Yes
Observations (firm-years)	1,866	21,425	2,071	19,532
Adjusted R^2	0.725	0.771	0.775	0.794

Panel B: Alternative benchmark samples				
Dependent variable =	*% women directors$_{t+1}$*			
Benchmark sample =	PSM U.S. firms	Industry- & size-matched U.S. firms	Non-U.S. firms	Canadian venture firms
	(1)	(2)	(3)	(4)
Post × Treat	**3.567*****	**4.451*****	**5.273*****	**3.876*****
	(5.08)	**(6.70)**	**(6.77)**	**(3.56)**
Controls	Yes	Yes	Yes	Yes
Firm & year fixed effects	Yes	Yes	Yes	Yes
Observations (firm-years)	3,837	4,824	3,892	2,568
Adjusted R^2	0.763	0.789	0.742	0.777

Panel C: Alternative dependent variables				
Dependent variable =	*N. women directors$_{t+1}$*	*Having women directors$_{t+1}$*	*N. men directors$_{t+1}$*	*N. directors$_{t+1}$*
Model =	Zero-inflated Poisson Model	Probit Model	Poisson Model	Poisson Model
	(1)	(2)	(3)	(4)
Post × Treat	**0.247*****	**0.695*****	**−0.045*****	**0.019**
	(3.82)	**(6.83)**	**(−5.30)**	**(1.59)**
Controls	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Observations (firm-years)	21,898	21,898	21,898	21,898

Notes. This table presents results of the robustness checks. Panel A reports results using alternative treatment samples and event windows, Panel B reports results using alternative benchmark samples, and Panel C reports results using alternative dependent variables. See Appendix C for variable definitions. The *t*-statistics reported in parentheses are calculated based on robust standard errors clustered by firm.

***, **, and * indicate significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Table 6, panel A, reports summary statistics on the additional variables. Panel B reports the regression results. In column (1), in which we focus on the treatment firms, the coefficient on *Post* is significantly positive, suggesting an increase in the likelihood of new women directors being appointed as a committee chair for the treatment firms following the regulation. In column (2), in which the EB U.S. firms are included, we find a significantly positive coefficient on *Post* × *Treat*, suggesting a greater increase among the treatment firms relative to the benchmark firms.[15] Whereas column (3) suggests a decreased likelihood of new women directors being appointed as the board chair or lead director among the treatment firms following the regulation, this decrease becomes insignificant when the EB U.S. firms are included in column (4). This result is consistent with board leadership positions being associated with longer tenure (Balsam et al. 2016).

In columns (1) and (2) of panel C, we find a mirror image of the result in panel B. New men directors among the treatment firms are less likely to be appointed as a committee chair following the regulation relative to the benchmark firms. We again find no evidence of changes in the likelihood of new men directors being appointed as the board chair or lead director after the regulation. These results suggest that the regulation has not only increased the presence of women but also enhanced their influence on boards.

Table 5. Analyses Conditional on Shareholder Oversight

Panel A: Summary statistics, year −1

Variable	Treatment sample (Obs. = 381 firms) Mean	Treatment sample (Obs. = 381 firms) Standard deviation	EB U.S. sample (Obs. = 3,026 firms) Mean	EB U.S. sample (Obs. = 3,026 firms) Standard deviation
UN PRI IO %	10.90	9.786	29.14	20.89
Pension & indp. IO %	21.69	18.22	47.82	32.65
Big 3 IO %	0.290	1.677	8.526	8.235
Underrep. of women dir.	0.265	0.442	0.487	0.500
TSX index	0.483	0.500	—	—
U.S. cross-listing	0.346	0.476	—	—

Panel B: Regression results conditional on proxies related to investor type

	Dependent variable = % *women directors*$_{t+1}$		
	Treatment & EB U.S. firms		
	(1)	(2)	(3)
Post × *Treat*	2.846***	3.025***	4.455***
	(3.42)	(3.54)	(7.03)
***Post* × *Treat* × *UN PRI IO %*$_{-1}$**	**0.127*****		
	(2.64)		
Post × *UN PRI IO %*$_{-1}$	0.038**		
	(2.21)		
***Post* × *Treat* × *Pension & indp. IO %*$_{-1}$**		**0.052****	
		(2.02)	
Post × *Pension & indp. IO %*$_{-1}$		0.019*	
		(1.74)	
***Post* × *Treat* × *Big 3 IO %*$_{-1}$**			**0.272****
			(2.00)
Post × *Big 3 IO %*$_{-1}$			0.075*
			(1.83)
Controls	Yes	Yes	Yes
Firm & year fixed effects	Yes	Yes	Yes
Observations (firm-years)	21,898	21,898	21,898
Adjusted R^2	0.784	0.783	0.781

Panel C: Regression results conditional on other proxies of shareholder oversight

	Dependent variable = % *women directors*$_{t+1}$		
	Treatment & EB U.S. firms (1)	Treatment firms only (2)	Treatment firms only (3)
Post × *Treat*	4.266***		
	(6.27)		
***Post* × *Treat* × *Underrep. of women dir.*$_{-1}$**	**2.259***		
	(1.95)		
Post × *Underrep. of women dir.*$_{-1}$	2.332***		
	(4.07)		
Post		3.227***	4.406***
		(4.08)	(6.11)
***Post* × *TSX index*$_{-1}$**		**3.523*****	
		(3.77)	
***Post* × *U.S. cross-listing*$_{-1}$**			**2.339****
			(2.30)
Controls	Yes	Yes	Yes
Firm & year fixed effects	Yes	Yes	Yes
Observations (firm-years)	21,898	2,339	2,339
Adjusted R^2	0.786	0.755	0.751

Notes. This table reports results of analyses conditional on the extent of shareholder oversight. Panel A presents descriptive statistics for the partitioning variables. Panels B and C present the regression results conditional on proxies related to investor type and other proxies of shareholder oversight. See Appendix C for variable definitions. The *t*-statistics reported in parentheses are calculated based on robust standard errors clustered by firm.

***, **, and * indicate significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Table 6. Gender Diversity Disclosure Regulation and Appointment of Key Board Positions

Panel A: Summary statistics

Variable	Treatment firms, %		EB U.S. firms, %	
	Preperiod (N = 496)	Postperiod (N = 612)	Preperiod (N = 3,930)	Postperiod (N = 4,385)
Woman committee chair, new directors	1.6	4.4	1.8	1.8
Woman board chair/lead director, new directors	0.4	0.2	0.2	0.3
Man committee chair, new directors	13.9	9.5	11.7	13.5
Man board chair/lead director, new directors	5.4	4.3	4.7	5.4

Panel B: Analyses of women appointed as committee chair or board chair/lead director among new directors

Dependent variable =	*Woman committee chair, new directors$_{t+1}$*		*Woman board chair/lead director, new directors$_{t+1}$*	
	Treatment firms only (1)	Treatment & EB U.S. firms (2)	Treatment firms only (3)	Treatment & EB U.S. firms (4)
Post	**0.585*****		**−0.940***	
	(2.91)		**(−1.87)**	
Post × Treat		**0.554*****		**−0.568**
		(2.71)		**(−1.40)**
Treat		−0.156		0.299
		(−0.88)		(1.00)
Controls	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	No	Yes	No	Yes
Observations (firm-years)	1,108	9,423	1,108	9,423
Pseudo R^2	0.126	0.154	0.377	0.155

Panel C: Analyses of men appointed as committee chair or board chair/lead director among new directors

Dependent variable =	*Man committee chair, new directors$_{t+1}$*		*Man board chair/lead director, new directors$_{t+1}$*	
	Treatment firms only (1)	Treatment & EB U.S. firms (2)	Treatment firms only (3)	Treatment & EB U.S. firms (4)
Post	**−0.243****		**−0.107**	
	(−2.19)		**(−0.71)**	
Post × Treat		**−0.313****		**−0.139**
		(−2.50)		**(−0.82)**
Treat		−0.013		0.042
		(−0.14)		(0.32)
Controls	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	No	Yes	No	Yes
Observations (firm-years)	1,108	9,423	1,108	9,423
Pseudo R^2	0.085	0.108	0.078	0.08

Notes. This table reports the results of the effect of Canada's gender diversity disclosure regulation on the likelihood of appointing a woman or man as chair of the four major committees or board chair/lead director. Panel A reports summary statistics of the dependent variables. Panels B and C report the regression results of woman (man) committee chair and woman (man) board chair/lead director, respectively. We restrict the analyses to firm-years with new directors. See Appendix C for variable definitions. The t-statistics reported in parentheses are calculated based on robust standard errors clustered by firm.

***, **, and * indicate significance at the 1%, 5%, and 10% two-tailed levels, respectively.

5.3. Diversity Commitments and Women's Representation on the Board

To shed light on the effectiveness of diversity commitments in binding firms' diversity progress, we examine whether a treatment firm's disclosed commitments are associated with a subsequent change in women's representation on the board following the regulation. We construct a commitment index, *DivCommit_Post*, by summing up the following four indicator variables: (1) *Termlimit_Post*, an indicator equal to one if a firm discloses the adoption of director term limits and other mechanisms of board renewal in a year; (2) *Policy_Post*, an indicator equal to one if a firm discloses the adoption of a written board-diversity policy in a year; (3) *Consider_Post*, an indicator equal to one if a firm discloses considerations of women's representation in the director identification and selection process in a year; and (4) *Target_Post*, an indicator equal to one if a firm discloses the adoption of targets regarding women's representation on the board in a year. We code each indicator as zero if a treatment firm discloses no adoption of a policy in a given year of the postregulation period, if a firm-year falls in the preregulation period, or if a firm is in the EB U.S. sample. Data on the gender diversity disclosure items are collected from the annual disclosure reviews of the regulation, which provide firm identity, disclosed information on the required items, and the associated explanations (available at the OSC website as discussed in Section 2.1).

Table 7, panel A, reports descriptive statistics of the diversity commitment index and the associated individual disclosure items for the treatment firms. Out of the 1,074 firm-years postregulation, 58%, 29.8%, 68.7%, and 14.8% disclose the adoption of director term limits and other mechanisms of board renewal (*Termlimit_Post*), a written board-diversity policy (*Policy_Post*), consideration of women's representation in director recruitment (*Consider_Post*), and a board gender-diversity target (*Target_Post*), respectively. The diversity commitment index's (*DivCommit_Post*) mean value is 1.713.

Table 7, panel B, presents regression results. In columns (1)–(6), we regress % *women directors* on *DivCommit_Post*, *Termlimit_Post*, *Policy_Post*, *Consider_Post*, *Target_Post*, and all four individual indicators, respectively, after including *Post* × *Treat* and the same control variables and fixed effects as in column (4) of Table 3. The coefficient on *Post* × *Treat* is insignificant at conventional levels in column (6), suggesting no difference in

Table 7. Diversity Commitments and Women Representation on Boards

Panel A: Descriptive statistics of the board diversity commitment index (*DivCommit_Post*) and individual items, Dec. 2014–Nov. 2018 (Observations = 1,074 firm-years)

Description and corresponding disclosure item in Form 58-101F1	Mean
DivCommit_Post, sum of four firm-year-level variables including	1.713
1. *Termlimit_Post*, an indicator equal to one if a firm discloses the adoption of director term limits and other mechanisms of board renewal in a year and equal to zero if a firm discloses no adoption.	0.580
2. *Policy_Post*, an indicator equal to one if a firm discloses the adoption of a written board diversity policy in a year and equal to zero if a firm discloses no adoption.	0.298
3. *Consider_Post*, an indicator equal to one if a firm discloses considerations of women representation in the director identification and selection process in a year and equal to zero if a firm discloses no consideration.	0.687
4. *Target_Post*, an indicator equal to one if a firm discloses the adoption of targets regarding women representation on the board in a year and equal to zero if a firm discloses no adoption.	0.148

Panel B: Regression results

	Dependent variable = % *women directors*$_{t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Post × *Treat*	1.315*	2.723***	3.423***	2.602***	3.810***	1.096
	(1.65)	(4.09)	(5.64)	(3.61)	(6.40)	(1.36)
DivCommit_Post	**1.657*** **					
	(5.03)					
Termlimit_Post		**2.515*** **				**2.117*** **
		(4.20)				**(3.59)**
Policy_Post			**2.789*** **			**1.888*** **
			(3.75)			**(2.63)**
Consider_Post				**2.400*** **		**1.684*** **
				(3.70)		**(2.73)**
Target_Post					**2.896** **	**0.707**
					(2.52)	**(0.61)**
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations (firm-years)	21,733	21,733	21,733	21,733	21,733	21,733
Adjusted R^2	0.791	0.788	0.788	0.788	0.787	0.791

Notes. Panel A reports descriptive statistics on the board diversity commitment indices and the individual items under Canada's gender diversity disclosure regulation. Panel B reports the results of the effect of the disclosure regulation on women directorships, conditional on the board diversity commitment indices and the individual items. The diversity commitment indices and the four individual indicators are lagged by one year, measured in the postregulation period and are set as zero for the preregulation treatment observations and the EB U.S. Benchmark firms. See Appendix C for variable definitions. The *t*-statistics reported in parentheses are calculated based on robust standard errors clustered by firm.

***, **, and * indicate significance at the 1%, 5%, and 10% two-tailed levels, respectively.

changes in women's representation on the board following the regulation between the EB U.S. Benchmark firms and treatment firm-years without disclosing any of the four diversity commitments.[16] The coefficients on the summary indicator (*DivCommit_Post*) and the four individual indicators are positive and significant, suggesting a greater postregulation increase in the percentage of women directors for firm-years with a stronger diversity commitment. In terms of economic significance, based on column (1), a one-unit increase in the overall diversity commitment for an average treatment firm is associated with a 21% increase in the percentage of women directors.[17] In column (6), in which all four individual indicators are included in one regression, the coefficients on all indicators except *Target_Post* remain significantly positive. Overall, the results in Table 7 suggest that the postregulation diversity commitments are associated with a subsequent increase in women's representation on the board.[18]

5.4. Diversity Commitments and Institutional Ownership

We next examine how a treatment firm's ownership held by ESG-friendly investors (i.e., signatories to the UN PRI, pension and independent institutional investors, and the Big Three investors) changes with its diversity commitments disclosed under the regulation. Whereas we expect ESG-friendly investors to respond favorably to firms' diversity commitments, some investors may adjust their investment more than others. In particular, we expect foreign institutional investors to respond more to the disclosures but not the Big Three investors. This is because, relative to domestic investors, foreign investors face greater costs in obtaining information about local companies and, therefore, should benefit more from disclosure regulations (DeFond et al. 2011). The Big Three investors, on the other hand, have relatively small holdings of Canadian stocks, and their influence on or interest in board gender-diversity issues among Canadian firms may be minimal.

Table 8, panel A, reports the descriptive statistics of the different types of institutional ownership, and panel B presents the regression results. To facilitate interpretation of the coefficients in the firm fixed effects models, we exclude *Inst. ownership* as a control variable. We add *All-men board* to control for poor diversity performance. The dependent variable in columns (1)–(4) is the ownership held by the signatories to the UN PRI, pension funds and independent institutions, foreign signatories of the UN PRI, and foreign pension funds and independent institutions, respectively. The coefficient on *DivCommit_Post* is positive across columns and significant except in column (2). These results suggest that foreign ESG-friendly institutional investors change their ownership in response to the disclosed diversity commitments. As expected, the coefficient on *DivCommit_Post* in column (5) is insignificant for the ownership held by the Big Three investors. Overall, these results are consistent with the view that firms anticipate increased shareholder oversight by adopting diversity policies to cater to prosocial investors and that these changes attract more investment from these investors ex post.[19]

5.5. Government Pressure as An Alternative Channel

Lastly, we investigate an alternative explanation that government pressure drives firms to increase board gender diversity. Whereas the Canadian government has issued statements supporting gender diversity at work, its support is mainly through mandated disclosures to facilitate investor engagement in the corporate sector. To shed light on this issue, we examine whether there is a larger increase in women's representation on the boards after the regulation for firms facing greater government pressure proxied by (1) a variable indicating whether a firm has at least one government contract in the preregulation period (*Canadian gov. contract*$_{pre}$) and (2) the proportion of revenue from government contracts in the preregulation period (*%Canadian gov. contract/Sales*$_{pre}$). We obtain the government procurement data from TenderAlpha. Table 9, panel A, reports the descriptive statistics of the two variables of government pressure, and panel B presents the regression results. The coefficients on the interaction terms between *Post* and the two government pressure variables are insignificant at conventional levels, suggesting that government pressure is unlikely a channel explaining our results.

6. Conclusions

We investigate the effect of Canada's 2014 regulation, which requires firms to disclose standardized information on policies related to board gender diversity. We argue that the required disclosures enable shareholders to be better monitors. Our results show that the percentage of women directors for firms listed on the TSX increases after the regulation relative to that for a benchmark group of U.S. firms. The cross-sectional analysis suggests that firms with stronger shareholder oversight experience greater improvement in board gender diversity. The increase in women's representation on boards after the regulation is substantive as reflected in more women being appointed as committee chairs. Greater disclosed commitments are associated with an increase in ownership of foreign ESG-friendly institutional investors.

Overall, our study contributes to the literature by documenting that standardized disclosure requirements of firms' board gender-diversity policies increase women's representation on corporate boards. Our study provides policy implications on the design of effective disclosure regulations by highlighting the importance of

Table 8. Diversity Commitments and Institutional Ownership

Panel A: Descriptive statistics

Ownership held by	Preperiod, N = 1,100 Mean	Preperiod, N = 1,100 Standard deviation	Postperiod, N = 1,074 Mean	Postperiod, N = 1,074 Standard deviation
UN PRI IO %	11.31	9.98	14.15	12.03
Pension & indp. IO %	24.84	19.85	29.14	22.09
Foreign UN PRI IO %	4.84	6.12	6.57	7.28
Foreign pension funds & indp. IO %	11.09	13.25	14.24	15.34
Big 3 IO %	0.61	2.50	0.42	1.04

Panel B: Regression results, treatment sample only

	Ownership held by				
Dependent variable =	*UN PRI IO* $\%_{t+1}$ (1)	*Pension & indp. IO* $\%_{t+1}$ (2)	*Foreign UN PRI IO* $\%_{t+1}$ (3)	*Foreign pension & indp. IO* $\%_{t+1}$ (4)	*Big 3 IO* $\%_{t+1}$ (5)
DivCommit_Post	**0.519***	**0.473**	**0.483****	**0.657***	**−0.112**
	(1.65)	**(0.92)**	**(2.19)**	**(1.82)**	**(−1.41)**
All-men board	−0.850	−1.651	−1.320**	−3.041**	0.015
	(−0.88)	(−1.10)	(−2.28)	(−2.59)	(0.10)
Woman CEO	1.520	−1.349	0.469	−1.851	−0.237
	(0.77)	(−0.38)	(0.22)	(−0.69)	(−0.84)
CEO-chair	1.344	3.050	0.199	1.140	−0.041
	(0.95)	(1.26)	(0.22)	(0.74)	(−0.17)
Indp. director	−1.240	−5.309	0.691	−3.005	−0.289
	(−0.38)	(−0.99)	(0.38)	(−0.75)	(−0.52)
Avg. dir. tenure	2.188*	4.751**	1.646*	2.656	−0.255
	(1.68)	(2.03)	(1.79)	(1.35)	(−1.11)
Board size	1.022	1.727	0.586	−1.710	−0.130
	(0.65)	(0.67)	(0.68)	(−0.71)	(−0.39)
Market-book	0.267**	0.619***	0.097	0.215*	0.019
	(2.10)	(2.73)	(1.16)	(1.75)	(1.06)
Leverage	−4.331	−8.095	−2.403	−5.297	−1.251
	(−1.47)	(−1.45)	(−1.11)	(−1.19)	(−1.64)
Firm size	2.008***	3.850***	0.884*	2.292***	0.202
	(3.05)	(3.43)	(1.70)	(2.62)	(1.59)
ROA	1.351	4.945**	−0.035	1.904	−0.247
	(1.12)	(2.32)	(−0.04)	(1.29)	(−0.76)
Ret	0.272	0.456	−0.071	−0.179	0.033
	(0.95)	(1.02)	(−0.42)	(−0.55)	(0.25)
Firm age	−5.019**	−8.375*	−3.372*	−3.316	−0.287
	(−2.02)	(−1.85)	(−1.88)	(−1.05)	(−0.56)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations (firm-years)	2,174	2,174	2,174	2,174	2,174
Adjusted R^2	0.761	0.806	0.762	0.806	0.424

Notes. This table reports results of the association between board diversity commitments and institutional ownership. Panel A presents descriptive statistics of different types of institutional ownership. Panel B reports the regression results of the association between diversity commitments and different types of institutional ownership. See Appendix C for variable definitions. The t-statistics reported in parentheses are calculated based on robust standard errors clustered by firm.

***, **, and * indicate significance at the 1%, 5%, and 10% two-tailed levels, respectively.

standardized reporting requirements in facilitating shareholder oversight and shaping firm behaviors. In 2018, a legislation in Canada required disclosures of diversity on corporate boards and senior management not only with respect to the representation of women, but also with respect to the representation of Aboriginal persons, persons with disabilities, and visible minorities. This legislation is generally viewed as a follow-up to the 2014 regulation (Jeffrey et al. 2019) and provides opportunities for future research to extend our findings to board-diversity disclosures beyond gender.

We caution, however, that we do not assess social externalities or compare the disclosure regulation with other regulatory measures, such as quota laws. Thus, our study does not speak to the overall welfare implication of the regulation, which remains a fruitful area for future

Table 9. Analysis Conditional on Government Contracts, Treatment Firms Only

Panel A: Summary statistics

	Treatment firms, N = 381	
Variable	Mean	Standard deviation
Canadian gov. contract$_{pre}$	0.0341	0.1818
%*Canadian gov. contract/Sales*$_{pre}$	0.0013	0.0072

Panel B: Regression results

	Dependent variable = % *women directors*$_{t+1}$	
	(1)	(2)
Post	5.148***	5.223***
	(8.50)	(8.60)
***Post* × *Canadian gov. contract*$_{pre}$**	**3.651**	
	(1.41)	
***Post* × %*Canadian gov. contract/Sales*$_{pre}$**		**68.098**
		(1.06)
Controls	Yes	Yes
Firm fixed effects	Yes	Yes
Observations (firm-years)	2,339	2,339
Adjusted R^2	0.749	0.749

Notes. This table reports results of the analysis of the effect of the disclosure regulation conditional on government pressure. Panel A presents descriptive statistics of the two variables of government pressure. Panel B reports the regression results. See Appendix C for variable definitions. The t-statistics reported in parentheses are calculated based on robust standard errors clustered by firm.

***, **, and * indicate significance at the 1%, 5%, and 10% two-tailed levels, respectively.

research. Nonetheless, our findings inform regulatory discussions by improving our understanding of the mechanism through which the disclosure regulations shape firm behaviors and ESG outcomes. Whereas our focus on shareholder oversight is motivated by the fact that shareholders are considered the most salient group that influences boards' decisions (Clarkson 1995, Gupta et al. 2022), we acknowledge that there may be other mechanisms through which the disclosure regulation affects board diversity (e.g., increased media coverage). Last but not least, as our sample firms are limited to those covered by BoardEx and larger companies take priority in BoardEx's coverage decisions (Fracassi and Tate 2012), our inferences may not generalize to midsized or smaller firms.

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Appendix A. Sample Disclosures Following Canada's 2014 Gender Diversity Disclosure Regulation (Emphasis Added)

A.1. Example of Commitments to Board Gender Diversity: Air Canada, Excerpts from the Proxy Circular for the Fiscal Year Ended on December 31, 2014

Diversity Policy

Board of Directors: The Board is committed to maintaining high standards of corporate governance in all aspects of Air Canada's business and affairs, and recognizes the benefits of fostering greater diversity, both in the boardroom and within our workforce in Canada. A fundamental belief of the Board is that a diversity of perspectives maximizes the effectiveness of the Board and decision-making in the best interests of the Corporation. This belief in diversity was confirmed in a written diversity policy adopted by the Board in February 2015.

The diversity policy states that candidates will be considered against objective criteria, having due regard to the benefits of diversity on the Board including gender. Accordingly, consideration of the number of women on the Board, along with consideration of whether other diverse attributes are sufficiently represented, is an important component in the search for and selection of candidates. When identifying potential candidates, the Nominating Committee will, in addition to its own search, strive to use resources of organizations advancing diversity in Canada or abroad, and where necessary, seek advice from independent search consultants.

The Board has established as its target that women represent at least 25% of the directors of Air Canada by 2017. Currently, one out of 10 directors (10%) is a woman, and following the shareholder meeting and assuming all director nominees are elected, two out of 11 directors (18%) will be women. The Nominating Committee will conduct periodic assessments to consider the level of women representation on the Board, and the Board members will also evaluate the effectiveness of the director selection and nomination process, including compliance with the diversity policy, through the Board's annual evaluation process.

…

The Board's goal is to be a balanced board comprised of members with diverse backgrounds, experience and tenure. In furtherance of that goal, the Board has implemented two primary mechanisms of Board renewal, namely, a retirement policy and an annual evaluation process, each of which are described below. The Board has not adopted term limits for directors, as the Board believes the retirement policy and the annual evaluation process are effective in achieving the appropriate level of renewal of the Board's membership.

A.2. Example of No Commitment to Board Gender Diversity: Microbix Biosystems Inc., Excerpts from the Proxy Circular for the Fiscal Year Ended on September 30, 2015

The Board has not appointed a nominating committee to appoint and assess directors, nor has it implemented a formal process for assessing directors. Appointments to the Board of Directors are discussed by the Board as a whole with a view to reflecting the interest of shareholders and the needs of the Company. Assessments of directors are conducted in the same way. New directors are provided with an orientation program consisting of informal meetings with other Board members, management, the Company's legal counsel and/or auditors, depending upon the new directors' appointments and wishes. The Board considers that the current size of the Board is appropriate for the Company's size, complexity and stage of development.

The Board has not adopted a written policy relating to the identification and nomination of women directors. Potential nominees for the Board are evaluated on the basis of experience, skill and ability and determining if the candidates' qualifications will meaningfully contribute to the effective functioning of the Board taking into consideration current Board composition and the skills and knowledge required to make the Board most effective.

The Board has not adopted a written policy relating to the identification and nomination of directors, including women directors. The Board believes that having written policies governing the selection of Board nominees could unduly restrict the Board's ability to select the most capable nominees that are free from conflicts of interest or other considerations that may impede the ability of a candidate to serve as a director of the Company.

The Board and management of the Company consist of a diverse set of individuals with a broad range of skill sets. At this time it does not have any women members and the Board has not adopted a specific target regarding women on the Board or in executive positions as candidates are selected based on the primary considerations of experience, skill and ability. The Company is an equal opportunity employer and candidates are thereby selected based on the primary considerations of experience, skill and ability. As such, the Company has not adopted a specific target regarding women on the Board or in executive positions.

As at the date hereof, no members of the Board are women and there is one woman executive officer of the Company, representing 14% of the 7 executive offices of the Company.

Appendix B. Excerpts of ISS Canada Proxy Voting Policy Update and Guidelines (Emphasis Added)

B.1. Excerpts from ISS Canada Policy—Director Elections—Board Gender Diversity (TSX-Listed Issuers), 2017

Background and Overview

Gender diversity has become a high-profile corporate governance concern in the Canadian market since updated disclosure requirements came into effect for the 2015 proxy season …

B.2. Excerpts from ISS Canadian Proxy Voting Guidelines for TSX-Listed Companies, Benchmark Policy Recommendations, Effective for Meetings on or After February 1, 2018

Vote case-by-case on director nominees, examining the following factors when disclosed:

› Independence of the board and key board committees;
› Disclosed commitment to board gender diversity;
› Attendance at board and committee meetings;
› Corporate governance provisions and takeover activity;
› Long-term company performance;
› Directors' ownership stake in the company;
› Compensation practices;
› Responsiveness to shareholder proposals; and
› Board accountability.

Gender Diversity Policy (S&P/TSX Composite Index companies only)

General Recommendation: For S&P/TSX Composite Index companies, generally vote withhold for the Chair of the Nominating Committee or Chair of the committee designated with the responsibility of a nominating committee, or Chair of the board of directors if no nominating committee has been identified or no chair of such committee has been identified, where:

› The company has not disclosed a formal written gender diversity policy*; and

› There are zero female directors on the board of directors.

**Per NI 58-101 and Form 58-101F1, the issuer should disclose whether it has adopted a written policy relating to the identification and nomination of women directors. The policy, if adopted, should provide a short summary of its objectives and key provisions; describe the measures taken to ensure that the policy has been effectively implemented; disclose annual and cumulative progress by the issuer in achieving the objectives of the policy, and whether and, if so, how the board or its nominating committee measures the effectiveness of the policy.*

Appendix C. Variable Definitions

Variable	Definition
Variables of interest	
% women directors	Percentage of women directors on the board
N. women directors	Number of women directors on the board
Having women directors	Indicator variable equal to one for any women directors on the board and zero otherwise
N. men directors	Number of men directors on the board
N. directors	Number of directors on the board
Woman (man) committee chair, new directors	Indicator variable equal to one if a firm appoints a woman (man) among new directors as chair on the four major committees (i.e., governance, audit, compensation, and nominating) in a year and zero otherwise.
Woman (man) board chair/lead director, new directors	Indicator variable equal to one if a firm appoints a woman (man) among new directors as the chair of the board or lead independent director in a year and zero otherwise
Treat	Indicator variable equal to one for Canadian TSX-listed firms and zero for benchmark firms.
Post	Indicator variable equal to one for the postperiod of December 2015 to November 2019 and zero for the preperiod of December 2010 to November 2014
Year −4 (*Year −3* or *Year −2*)	Indicator variable equal to one for the fourth (third or second) year prior to December 2014 and zero otherwise
Year 1 (*Year 2*, *Year 3*, or *Year 4*)	Indicator variable equal to one for the first (second, third, or fourth) year after November 2015 and zero otherwise.
DivCommit_Post	Sum of the following four *firm-year-level* disclosure indicators (*Termlimit_Post*, *Policy_Post*, *Consider_Post*, and *Target_Post*)
ΔDivCommit_Post	*DivCommit_Post* in year $t + 1$ minus *DivCommit_Post* in year t in the postregulation period
Termlimit_Post	Indicator variable equal to one if a firm discloses the adoption of director term limits and other mechanisms of board renewal in a year and equal to zero if a firm discloses no adoption
Policy_Post	Indicator variable equal to one if a firm discloses the adoption of a written board-diversity policy in a year and equal to zero if a firm discloses no adoption
Consider_Post	Indicator variable equal to one if a firm discloses considerations of women's representation in the director identification and selection process in a year and equal to zero if a firm discloses no consideration
Target_Post	Indicator variable equal to one if a firm discloses the adoption of targets regarding women's representation on the board in a year and equal to zero if a firm discloses no adoption
Firm-level controls	
Inst. ownership	Total number of shares owned by institutional investors divided by sharesoutstanding
Woman CEO	Indicator variable equal to one if the firm has a woman CEO and zero otherwise
CEO-chair	Indicator variable equal to one if a firm's CEO is the chairperson of the board and zero otherwise
Indp. directors	Proportion of independent directors
Avg. dir. tenure	Natural logarithm of one plus the average tenure of the firm's directors
Board size	Natural logarithm of the number of directors
Market-book	Ratio of the market value of equity to the book value of equity
Leverage	Short-term debt plus long-term debt divided by total assets
Firm size	Natural logarithm of total assets in millions of U.S. dollars
ROA	Pretax income divided by total assets
Ret.	Annual stock returns adjusted by the two-digit Standard Industrial Classification industry return for Canadian and U.S. firms and by the market return for non-U.S. firms
Firm age	Natural logarithm of the age of the firm, measured as the number of years since the firm was incorporated or founded, whichever earlier. When such dates are unavailable, firm age is calculated as the number of years since the firm first appeared in Worldscope or Compustat.
Additional variables	
UN PRI IO %	Percentage of ownership held by institutions that are signatories to the UN PRI
Pension & indp. IO %	Percentage of ownership held by pension funds and independent institutions (i.e., mutual funds and investment advisers)
Big 3 IO %	Percentage of ownership held by the Big Three institutional investors (i.e., Blackrock, State Street Global Advisors, and Vanguard)
Underrep. of women dir.	Indicator variable equal to one if the number of women directors is less than the industry median and zero otherwise
TSX Index	Indicator variable equal to one if a firm is included in the TSX composite index and zero otherwise
U.S. cross-listing	Indicator variable equal to one if a firm is cross-listed in a U.S. stock exchange and zero otherwise
All-men board	Indicator variable equal to one if a firm has no woman director and zero otherwise
Foreign UN PRI IO %	Percentage of ownership held by foreign institutions that are signatories to the UN PRI
Foreign pension & indp. IO %	Percentage of ownership held by foreign pension funds and independent institutions
Canadian gov. contract$_{pre}$	Indicator variable that equals one if the treatment firm has at least one contract with the Canadian government in the preperiod and zero otherwise
%Canadian gov. contract/Sales$_{pre}$	One hundred times the contract dollar value awarded by the Canadian government in a year divided by the firm's annual sales, summed over the four preperiod years

Endnotes

[1] We define standardized disclosures as disclosures that meet four principles: (1) clear, (2) meaningful, (3) consistent over time, and (4) comparable across firms (see https://www.bankinghub.eu/finance-risk/standardization-disclosure-active-management).

[2] Firms may voluntarily disclose their diversity policies to signal their performance (Lys et al. 2015). Theory suggests that voluntary disclosure may not serve as an effective commitment because firms can cease to provide the information or switch to generic discussions whenever they see fit (Cheng et al. 2013). Consistent with this observation, we find that voluntary diversity disclosures tend to be spotty, vague, and vary in scope.

[3] A related stream of literature investigates the association between gender-diverse boards and various outcomes, such as firm value, stock price informativeness, reporting quality, governance quality, and mergers and acquisition (e.g., Adams and Ferreira 2009, Gul et al. 2011, Srinidhi et al. 2011, Levi et al. 2014, Cumming et al. 2015, Chen et al. 2016).

[4] Several states in the United States have recently enacted disclosure mandates on boardroom gender diversity (e.g., Maryland and Illinois in 2019). Lu (2019) compares the effects of quota laws and disclosure-related governance reforms on the characteristics of new women directors. Different from our study, Lu (2019) focuses on regulations in Europe, a setting in which the disclosure requirements are often bundled with governance code reforms (Fauver et al. 2024).

[5] National Instruments are the set of rules and regulations developed and implemented by the CSA, which consists of securities regulators in Canada's 10 provinces and three territories and is an umbrella organization that coordinates and harmonizes the securities regulations across Canada.

[6] For the full text, see https://www.osc.ca/en/securities-law/instruments-rules-policies/5/58-101/amendment-instrument-ni-58-101-disclosure-corporate-governance-practices.

[7] The reports and data are provided in the category of ongoing requirements for issuers and insiders (category 5) in instruments, rules, and policies (e.g., 2015 review available at https://www.osc.ca/en/securities-law/instruments-rules-policies/5/58-307).

[8] In additional analyses (untabulated), we find that both the EB U.S. firms and Canadian firms experience few changes and exhibit similar trends in women's representation on the board in the four years surrounding the adoption of the 2009 SEC diversity disclosure rule.

[9] A written board-diversity policy is a key item of the disclosure regulation. The investor states, "If Restaurant Brands International is going to change this picture, it needs both a policy and a plan" (Middlemiss 2016). We note, however, that shareholder proposals are rare in Canada (Insightia 2022).

[10] The use of such a sample (referred to as a balanced sample) is in line with prior studies that use a DiD design (e.g., Tan et al. 2011, Hung et al. 2015). In Online Table A1, we use a constant sample that requires firms to have necessary data in each year and find similar results. We do not use the constant sample, which comprises 1,016 firm-years for 127 unique firms, for our main analyses because it lacks representation of the TSX population and is relatively prone to a survivorship bias.

[11] Additional analyses (untabulated) find that our results are robust to alternative clustering schemes at the industry or year level or two-way clustering at the firm and year levels or the industry and year levels.

[12] Fifty-three percent = 4.2/7.88, where 4.2 is the coefficient on *Post* × *Treat* in column (4) of Table 3 and 7.88 is the average percentage of women directors for the treatment firms in the preregulation period as reported in panel B of Table 2; 1.33 = 0.869 × (1 + 53%), where 0.869 is the average number of women directors for the treatment firms in the preregulation period, and 53% is the relative increase in percentage.

[13] Firms may have some policies on diversity, equity, and inclusion internally, but the policy is not specific to the board recruitment process. Based on a 2013 OSC survey to TSX-listed issuers, "91% of respondents do not have a policy for the identification and nomination of women directors" (Ontario Securities Commission 2014, p. 8). This survey suggests that a vast majority of TSX firms did not have a robust board gender-diversity policy in place prior to the regulation.

[14] By "our results remain robust," we mean that the coefficient on *Post* using the model in column (2) of Table 3 and the coefficient on *Treat* × *Post* using the model in column (4) of Table 3 are positive and significant at $p \leq 0.10$, two-tailed.

[15] Additional analysis (untabulated) finds that both the percentage of women among newly appointed directors and the percentage of women directors on major committees increase after the regulation among the treatment firms relative to the EB U.S. firms. In contrast, in Online Tables A6 and A7, we find no evidence of changes in other governance measures (i.e., percentage of independent directors and CEO-chair duality) and other forms of board diversity (i.e., visible minorities, measured as the percentage and likelihood of board members from countries outside of North America, Europe, Australia, and New Zealand).

[16] In Online Table A8, we perform an analysis conditional on whether a firm discloses any commitment in the postregulation period and find results consistent with this inference. Only 12% of the treatment firms (40 firms) do not disclose any commitment postregulation, and there is no significant difference in postregulation changes in women directorships between these firms and the EB U.S. benchmark firms.

[17] Twenty-one percent = 1.657/7.88, where 1.657 is the coefficient on *DivCommit_Post* in column (1) and 7.88 is the average percentage of women directors for the treatment firms in the preregulation period as reported in panel B of Table 2.

[18] In Online Figure A1, we plot the coefficients on the interactions between *High DivCommit* and the year indicators estimated by regressions that include the control variables used in the main analyses and year and firm fixed effects, for which *High DivCommit* is an indicator equal to one for firms with diversity commitment in year *1* above the sample median and zero otherwise. We find that, whereas there is a slight upward trend for firms with high diversity commitments (i.e., *High DivCommit* = 1), these firms increase the percentage of women directors more relative to those with low commitments after the regulation. We also validate the parallel trends assumption for firms with zero versus nonzero diversity commitments in year 1 (untabulated).

[19] In Online Table A9, we also find that, after the regulation, firms that make stronger diversity commitments gain greater shareholder voting support in director elections. In addition, firms receiving more unfavorable votes in director elections increase diversity commitments postregulation, suggesting that some firms may fail to anticipate (or underestimate) the extent of increased shareholder oversight.

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